

Intermediate Macro

Chapter 12: Technological Progress and Growth

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Figure 1

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Abstract

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Technical Progress

12.1 Technological Progress and the Rate of Growth

- Technical progress has many dimensions:
 - It can lead to larger quantities of output for given quantities of capital and labor (e.g. a new type of lubricant).
 - It can lead to better products (e.g. steady improvement in automobile safety and comfort).
 - It can lead to new products (e.g. iPad, wireless communication technology, flat screen monitors, and self-driving cars).
 - It can lead to a larger variety of products (e.g. steady increase in the number of breakfast cereals available at your local supermarket).
- If we think of output as the underlying services provided by the goods produced in the economy, we can think of technological progress as leading to increases in output for given amounts of capital and labor.

An Extended Production Function

12.1 Technological Progress and the Rate of Growth

- We can think of **state of technology** as a variable that tells us how much output can be produced from given amounts of capital and labor at any time.
- If we denote the state of technology by A , we can rewrite the production function as

$$Y = F(K, N, A)$$

(+, +, +)

- Incorporate labor-augmenting technology into the aggregate production function:

$$Y = F(K, AN) \tag{12.1}$$

- We can think of AN as the amount of **effective labor** in the economy.

Relation between $\frac{Y}{AN}$ and $\frac{K}{AN}$

12.1 Technological Progress and the Rate of Growth

- The function F is assumed to exhibit constant returns to scale and therefore decreasing returns to each of its arguments separately.
- To get a relation between output per effective worker and capital per effective worker, dividing both sides by AN in the equation (12.1) gives

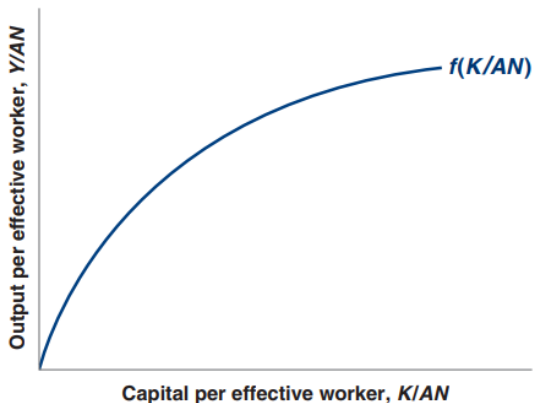
$$\tilde{y} = \frac{Y}{AN} = F\left(\frac{K}{AN}, 1\right) = F(\tilde{k}, 1)$$

- Or, if we define the function f so that $f(\tilde{k}) = F(\tilde{k}, 1)$:

$$\tilde{y} = f(\tilde{k}) \tag{12.2}$$

$$\frac{Y}{AN} \text{ versus } \frac{K}{AN}$$

12.1 Technological Progress and the Rate of Growth



Interactions between Output and Capital

12.1 Technological Progress and the Rate of Growth

- With the public saving set equal to zero, goods market equilibrium in a closed economy is equivalent to

$$I = S = sY$$

- Divide both sides by the number of effective workers, AN , to get

$$\frac{I}{AN} = s \frac{Y}{AN}$$

- Replacing output per effective worker by its expression from equation (12.2) gives

$$\iota = \frac{I}{AN} = sf(\tilde{k})$$

Steady-State Equilibrium Condition of i

12.1 Technological Progress and the Rate of Growth

- Let δ be the depreciation rate of capital. Let the rate of technological progress be equal to g_A . Let the rate of population growth be equal to g_N .
- These assumptions imply that the level of investment needed to maintain a given level of capital per effective worker is therefore given by

$$I = \delta K + (g_A + g_N)K$$

- Or, equivalently,

$$I = (\delta + g_A + g_N)K \quad (12.3)$$

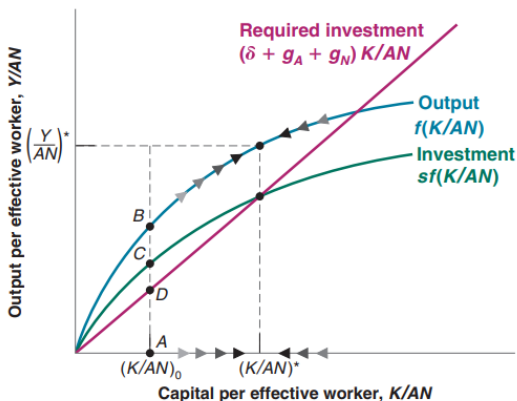
- Dividing both sides by AN gives

$$\iota = \frac{I}{AN} = (\delta + g_A + g_N)\tilde{k}$$

Dynamics of Capital and Output

12.1 Technological Progress and the Rate of Growth

- We can now give a graphical description of the dynamics of $\tilde{k}$ and $\tilde{y}$.



Balanced Growth

12.1 Technological Progress and the Rate of Growth

- In the long run, capital per effective worker reaches a constant level, and so does output per effective worker.
- Put another way, the steady state of the economy is such that *capital per effective worker and output per effective worker are constant and equal to $\tilde{k}^*$ and $\tilde{y}^*$, respectively.*
- This implies that, in steady state, output Y is growing at the same rate as effective labor AN .
- Output growth in steady state must equal $g_A + g_N$. The same reasoning applies to capital. In steady state, capital stock K is also growing at rate $g_A + g_N$.
- Because output, capital, and effective labor all grow *at the same rate* $g_A + g_N$ in steady state, the steady state of this economy is also called a state of **balanced growth**.

The Characteristics of Balanced Growth

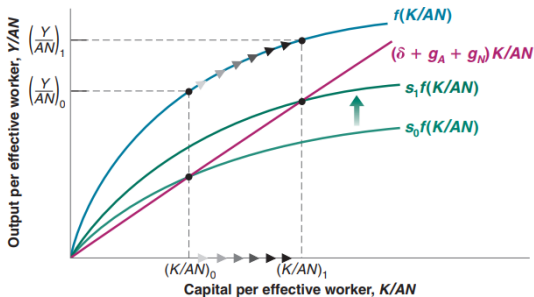
12.1 Technological Progress and the Rate of Growth

Table 12-1 The Characteristics of Balanced Growth		
		Growth Rate
1	Capital per effective worker	0
2	Output per effective worker	0
3	Capital per worker	g_A
4	Output per worker	g_A
5	Labor	g_N
6	Capital	$g_A + g_N$
7	Output	$g_A + g_N$

Back to the Effects of the Saving Rate

12.1 Technological Progress and the Rate of Growth

- In steady state, the growth rate of output depends *only* on the rate of population growth and the rate of technological progress.
- Changes in the saving rate do not affect the steady-state *growth rate*, but do increase the steady-state *level* of output per effective worker.



12.1 Technological Progress and the Rate of Growth



Sources of Technological Progress

12.2 The Determinants of Technological Progress

- The model in this chapter establishes that long-run growth is determined by the rate of technological progress, but takes this rate as given.
- What are the sources of technological progress?
- Technological change is the product of **research and development (R&D)**, most of which is conducted by firms in search of increased profits.
- In general terms, R&D spending depends on
 - ① the expected **fertility of research** (its yield of new ideas and products) and
 - ② the **appropriability** of the results of research.

The Fertility of the Research Process

12.2 The Determinants of Technological Progress

- Fertility depends on the successful interaction of basic research, applied research, and product development.
- Some countries appear more successful at basic research; other countries are more successful at applied research and development.
- Studies point to differences in the education system as one of the reason why.
- An age-old worry is that research will become less and less fertile, that most major discoveries have already taken place and that technological progress will begin to slow down.
- But so far there is no evidence that it is correct.

The Appropriability of Research Results

12.2 The Determinants of Technological Progress

- Appropriability depends on in part on the nature of the research process.
- If it is believed that a new discovery will quickly lead to a better discovery by firm, the ability to profit from the results of the research is limited.
- Appropriability also depends on the degree of **patent protection** afforded to the inventors of new products.
- Patent protection allows the inventor of a product to enjoy a monopoly on its sale for a time, and thus offers an incentive for research effort.
- On the other hand, patent protection makes it more expensive for society to benefit from the introduction of new products, since these products are likely to be sold at prices above their marginal cost of production.

Innovation versus Imitation

12.2 The Determinants of Technological Progress

- Although R&D is clearly central to technological progress, other dimensions are also relevant.
- Existing technologies can be used more or less efficiently.
- Good management makes a substantial difference to the productivity of firms.
- For some countries, R&D may be less important than for others.
- To sustain growth, advanced countries, which are at the **technology frontier**, must innovate.
- Poorer countries, which are further from the technology frontier, can instead grow largely by imitating rather than innovating, by importing and adapting existing technologies instead of developing new ones.

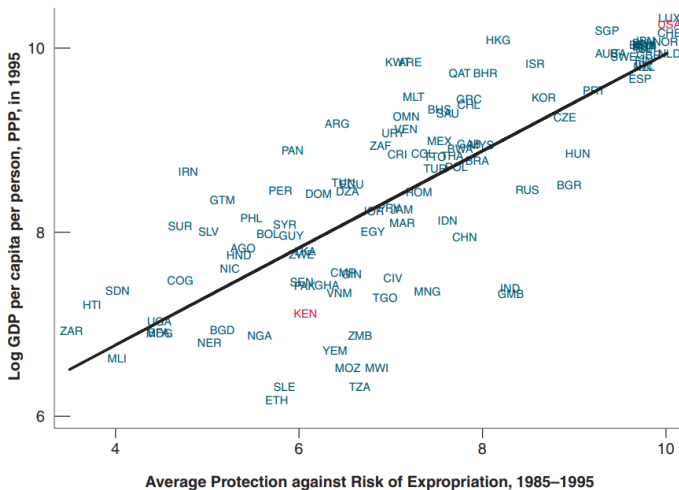
Property Rights

12.3 Institutions, Technological Progress, and Growth

- Growth rates differ significantly among countries.
- While many factors contribute to these differences, many economists increasingly point to institutions as the fundamental reason why many poor countries remain poor.
- By institutions, economists generally mean the procedures and tradition in place to protect private **property rights**.
- These procedures and traditions span a wide range, including areas as diverse as the degree of corruption in the political system, the efficiency and fairness of the court system, and patent and anti-trust law and enforcement, among many others.
- The complexity and scope of the arrangements required to protect property rights in a modern economy make it difficult for poor countries to build these institutions.

Protection from Expropriation and GDP per Person

12.3 Institutions, Technological Progress, and Growth



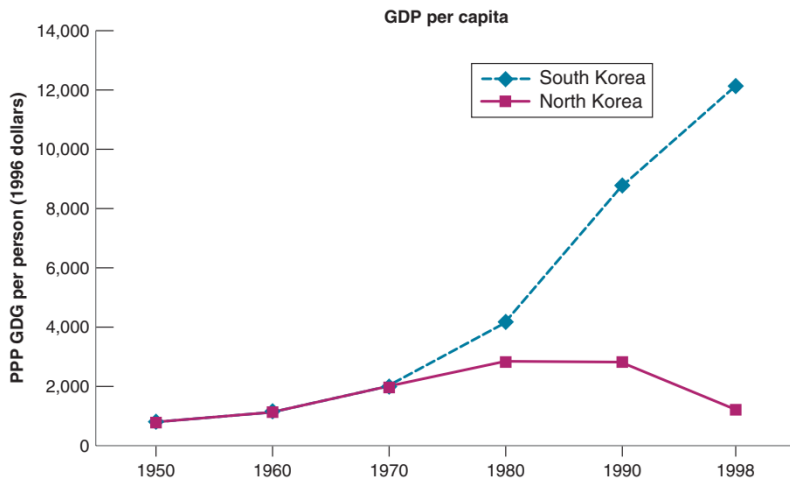
Case Study in Property Rights

12.3 Institutions, Technological Progress, and Growth

- The Importance of Institutions: North Korea and South Korea
- Prior to 1953, Korea was a single country with no physical differences in population or resources, in other words a very homogeneous country.
- After 1953, the country was divided into North Korea (limited property rights and central planning) and South Korea (protection of property rights and capitalism).
- A substantial divergence in per capita GDP now exists after more than 60 years.
- The difference is due to differences in institutions.

PPP GDP per Person: North and South Korea

12.3 Institutions, Technological Progress, and Growth



How to Measure Technological Progress

Appendix

- How do we measure the rate of technological progress?
- The answer was given in 1957 by Robert Solow and is still in use today.
- It relies on one important assumption that each factor of production is paid its marginal product.
- Under this assumption, it is easy to compute the contribution of an increase in any factor of production to the increase in output.
- Assume that the aggregate production function is of following form

$$Y = (AN)^{\alpha} K^{1-\alpha}$$

Solow Residual

Appendix

- We can then measure the effects of technological progress by computing what Solow called the residual, the excess of actual growth of output g_Y over the growth attributable to growth of labor and the growth of capital.

$$residual \equiv g_Y - [\alpha g_N + (1 - \alpha)g_K]$$

- The Solow residual is sometimes called the **rate of growth of total factor productivity** (or the **rate of TFP growth**).
- The residual is equal to the share of labor times the rate of technological progress:

$$residual = \alpha g_A$$

What Lies Behind Chinese Growth?

Appendix

- During the period of high growth, output growth was very high, 7.2%. Growth of output per worker was also very high, at 5.5%.
- The rate of technological progress was also high, but not quite as much, namely 4.2%.
- The difference between the growth rate of output and the rate of technological progress reflected a steady increase in the ratio of capital to output, equal to 2% a year.
- Eventually, the ratio of capital to output will stabilize, and the growth of output per worker will reflect the rate of technological progress.
- Thus, if this rate remains the same, China's growth of output per worker will decrease to 4.2%.

Application to China

Appendix

- If growth of labor remains the same, namely, 1.7%, then growth of output will converge to 5.9%.
- This does indeed roughly correspond to forecasts of future growth in China.